UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	NOVEMBER
TIME	2026

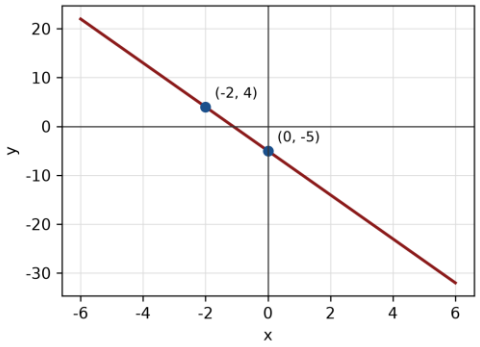
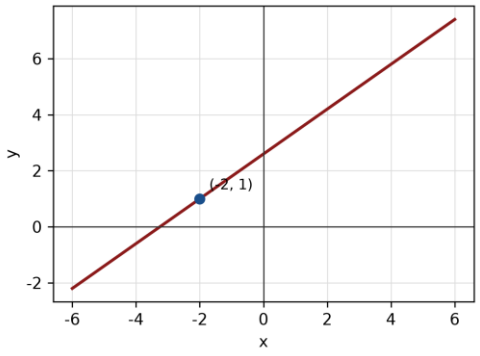
INSTRUCTIONS TO MARKERS

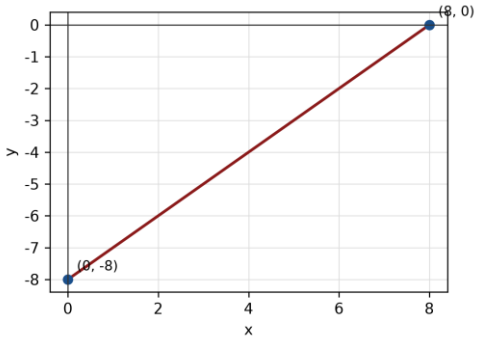
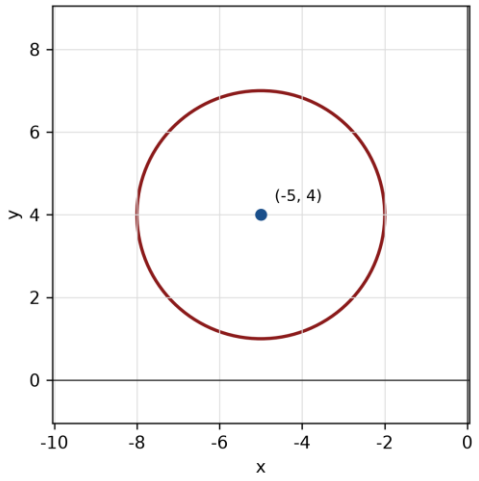
1. This marking scheme accompanies the MAT133 mock paper (November 2026).
2. Answers and marking points are for local study use only.

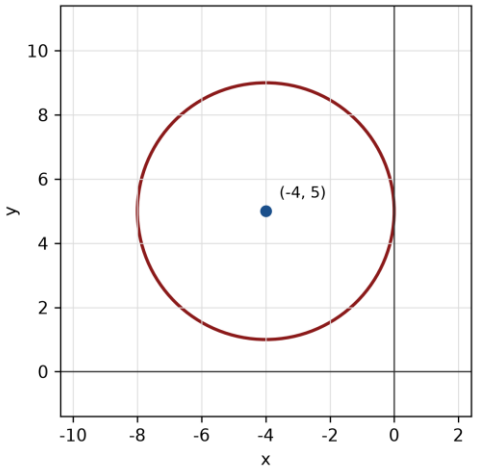
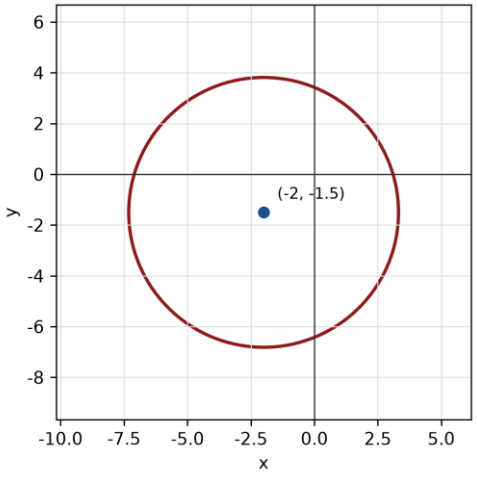
FOR MARKERS' USE ONLY

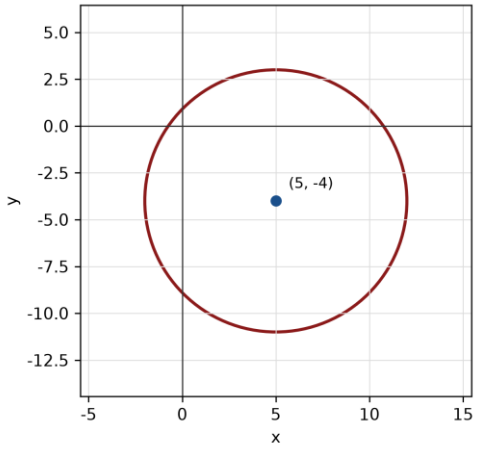
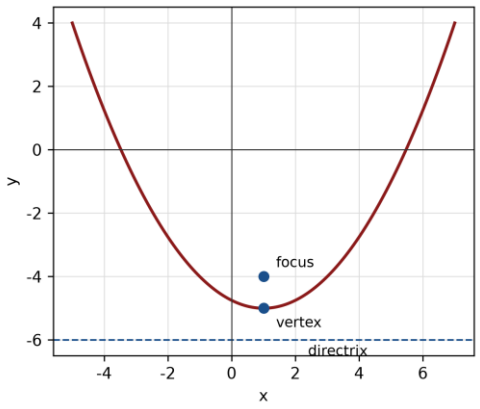
This marking scheme consists of 6 printed pages

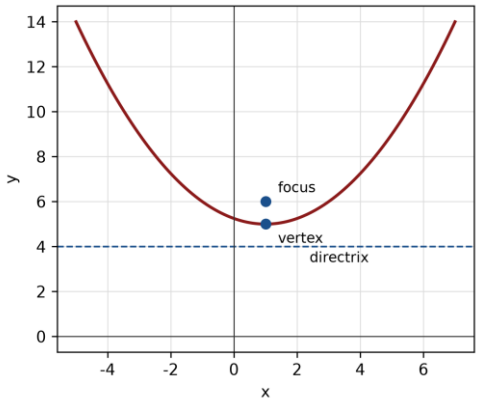
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NO	SOLUTION	MARK
1 a)	$m = -\frac{9}{2}$ M1 $y + 5 = -\frac{9x}{2}$ M1 $y = -\frac{9x}{2} - 5$ A1 	4 m
1 b)	$-4x + 5y + 3 = 0$ M1 $y = \frac{4x}{5} - \frac{3}{5}$ M1 $m_{\text{new}} = \frac{4}{5}$ $y = \frac{4x}{5} + \frac{13}{5}$ A1 	5 m

NO	SOLUTION	MARK
1 c)	$\frac{x}{8} - \frac{y}{8} = 1$ M1 $-8x + 8y = -64$ $y = x - 8$ A1 	4 m
2 a)	$(x+5)^2 + (y-4)^2 = 9$ M1 $x^2 + 10x + y^2 - 8y + 41 = 9$ A1 $x^2 + 10x + y^2 - 8y + 32 = 0$ B1 	5 m

NO	SOLUTION	MARK
2 b)	$x^2 + 8x + y^2 - 10y = -25$ M1 M1 $(x+4)^2 + (y-5)^2 - 41 = -25$ A1 $(x+4)^2 + (y-5)^2 = 16$ A1 	5 m
2 c)	$\left(-2, -\frac{3}{2}\right)$ M1 $r^2 = \frac{113}{4}$ M1 $(x+2)^2 + \left(y+\frac{3}{2}\right)^2 = \frac{113}{4}$ A1 	5 m

NO	SOLUTION	MARK
2 d)	$(x-5)^2+(y+4)^2=49$ M1 A1 centre $(5,-4)$, $r=7$ A1 	4 m
3 a)	$y=\frac{x^2}{4}-\frac{x}{2}-\frac{19}{4}$ M1 A1 $y=\frac{(x-1)^2}{4}-5$ A1 $(x-1)^2=4y+20$ A1 	5 m

NO	SOLUTION		MARK
3 b)	vertex $(1, 5)$	M1	5 m
	$p = 1$	M1	
	$(x-1)^2 = 4y - 20$	A1	
			
3 c)	$(x+3)^2 = 4y - 20$	A1	5 m
	vertex $(-3, 5)$	A1	
	focus $(-3, 6)$	A1	
	directrix $y = 4$	B1	
	